

Individual Assignment 5 JM

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Set Up

Packages

```
# general and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
```

```

# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

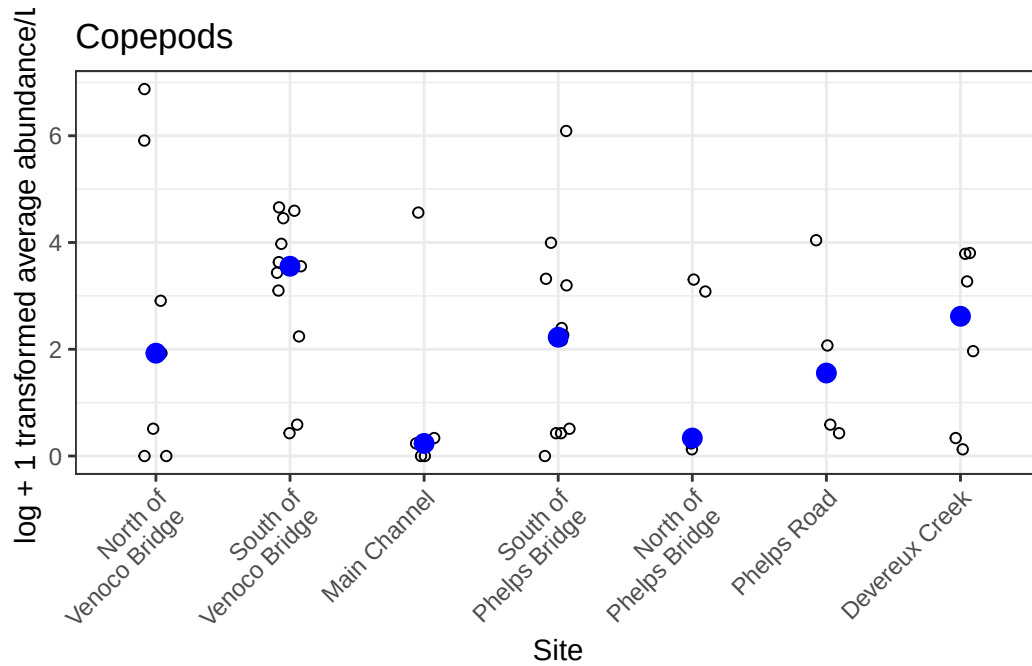
```

```
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
  site_full = fct_rev(site_full))
```

```
# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))
```

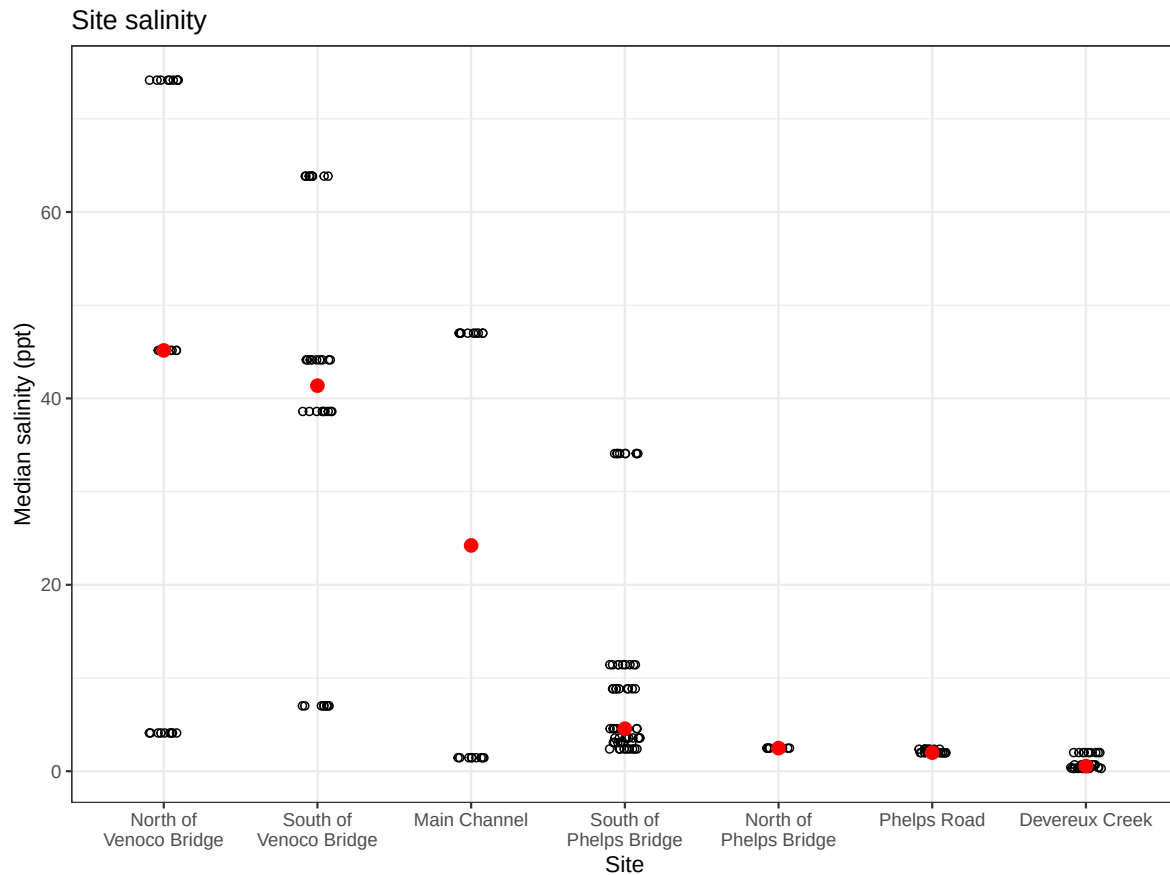
```
copepods_plot <- ggplot(data = copepods, mapping = aes(x = site_full, y = log_ave_lit)) +
  geom_jitter(height = 0, shape = 21, width = 0.1) +
  stat_summary(geom = "point", fun = "median", size = 3, color = "blue") +
  scale_x_discrete(labels = label_wrap(15)) +
  labs(x = "Site", y = "log + 1 transformed average abundance/L", title = "Copepods") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

copepods_plot
```



```
salinity <- ggplot(data = aq_ins_clean,
  mapping = aes(x = site_full,
    y = med_sal)) +
  geom_jitter(height = 0,
    width = 0.1,
    shape = 21) +
  scale_x_discrete(labels = label_wrap(width = 15)) +
  stat_summary(fun = median,
    color = "red",
    size = 0.5) +
  labs(x = "Site",
    y = "Median salinity (ppt)",
    title = "Site salinity") +
  theme_bw()
```

salinity



Problems

1. Visualizing differences across sites in major taxon abundance

a. Planning figure

- X axis: site (site_full)
- Y axis: ostracod abundance (log + 1 transform)
- geometry to represent average abundance/L: geom_jitter()
- geometry to represent medians: stat_summary()

b. Wrangle data

```
ostracods <- aq_ins_clean |>
  # filtering to only include ostracod species
  filter(taxon == "ostracod") |>
  # transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
# show random sample of 10
slice_sample(ostracods, n = 10)
```

A tibble: 10 x 24

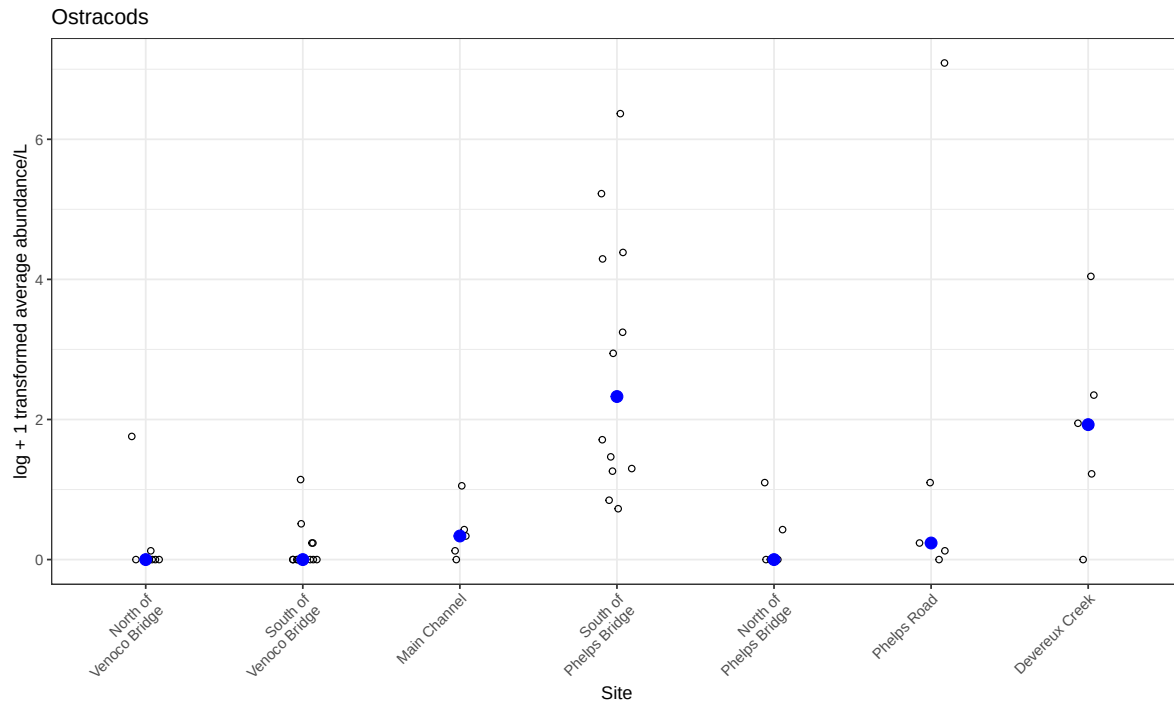
	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
2	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	ostr~	0	NA	NA	NA
3	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.13	NA	NA	NA
4	2023-05-04_NMC	2023-05-04 00:00:00	NMC	ostr~	1.87	8.32	NA	10.3
5	2022-08-12_NEC	2022-08-12 00:00:00	NEC	ostr~	0	7.37	NA	NA
6	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	ostr~	0	NA	NA	NA
7	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38
8	2022-11-19_NDC	2022-11-19 00:00:00	NDC	ostr~	6	NA	NA	NA
9	2021-11-12_NPB	2021-11-12 00:00:00	NPB	ostr~	79.2	NA	NA	NA
10	2022-01-28_NDC	2022-01-28 00:00:00	NDC	ostr~	9.47	7.65	2.00	9.55

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
comments <chr>, site_full <fct>, log_ave_lit <dbl>

c. Coding figure

```
ostracod_plot <- ggplot(data = ostracods, mapping = aes(x = site_full, y = log_ave_lit)) +
  geom_jitter(height = 0, shape = 21, width = 0.1) +
  stat_summary(geom = "point", fun = "median", size = 3, color = "blue") +
  scale_x_discrete(labels = label_wrap(15)) +
  labs(x = "Site", y = "log + 1 transformed average abundance/L", title = "Ostracods") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

ostracod_plot
```



d. Adding ostracod panel

```
salinity / (copepods_plot | ostracod_plot)
```


Scatter plot showing the relationship between Site and the number of fish caught. The x-axis lists seven sites: North of Venoco Bridge, South of Venoco Bridge, Main Channel, South of Phelps Bridge, North of Phelps Bridge, Phelps Road, and Devereux Creek. The y-axis represents the number of fish caught, ranging from 0 to 10. Data points are represented by open circles (O) and solid red circles (●).

Site	Fish Caught (O)	Fish Caught (●)
North of Venoco Bridge	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3, 4, 5, 6, 7, 8, 9, 10
South of Venoco Bridge	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3, 4, 5, 6, 7, 8, 9, 10
Main Channel	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3, 4, 5, 6, 7, 8, 9, 10
South of Phelps Bridge	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3, 4, 5, 6, 7, 8, 9, 10
North of Phelps Bridge	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3, 4, 5, 6, 7, 8, 9, 10
Phelps Road	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3, 4, 5, 6, 7, 8, 9, 10
Devereux Creek	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	3, 4, 5, 6, 7, 8, 9, 10

Scatter plot showing the log + 1 transformed average abundance/L for various sites. The y-axis ranges from 0 to 6. The x-axis lists seven sites: North of Venoco bridge, South of Venoco Bridge, Main Channel, South of Phelps bridge, North of Phelps bridge, Phelps Road, and Devereux Creek. Data points are shown as open circles, with some points highlighted in blue.

Site	log + 1 transformed average abundance/L	Highlighted
North of Venoco bridge	0.0	No
North of Venoco bridge	0.0	No
North of Venoco bridge	0.5	No
North of Venoco bridge	2.9	No
North of Venoco bridge	5.9	No
North of Venoco bridge	6.8	No
South of Venoco Bridge	0.4	No
South of Venoco Bridge	0.5	No
South of Venoco Bridge	2.2	No
South of Venoco Bridge	3.1	No
South of Venoco Bridge	3.5	No
South of Venoco Bridge	3.6	No
South of Venoco Bridge	4.0	No
South of Venoco Bridge	4.5	No
South of Venoco Bridge	4.6	No
Main Channel	0.0	No
Main Channel	0.0	No
Main Channel	0.2	No
Main Channel	0.3	No
Main Channel	4.5	No
South of Phelps bridge	0.0	No
South of Phelps bridge	0.4	No
South of Phelps bridge	0.4	No
South of Phelps bridge	2.2	No
South of Phelps bridge	2.3	No
South of Phelps bridge	2.4	No
South of Phelps bridge	3.2	No
South of Phelps bridge	3.3	No
South of Phelps bridge	4.0	No
South of Phelps bridge	6.1	No
North of Phelps bridge	0.2	No
North of Phelps bridge	0.3	No
North of Phelps bridge	3.1	No
North of Phelps bridge	3.3	No
Phelps Road	0.4	No
Phelps Road	0.5	No
Phelps Road	1.5	No
Phelps Road	2.1	No
Phelps Road	4.0	No
Devereux Creek	0.1	No
Devereux Creek	0.3	No
Devereux Creek	2.6	No
Devereux Creek	3.3	No
Devereux Creek	3.8	No
Devereux Creek	3.8	No
Devereux Creek	2.6	Yes
Devereux Creek	2.0	No

Scatter plot showing the log + 1 transformed average abundance/L for various sites. The y-axis ranges from 0 to 6. The x-axis lists sites: North of Venoco bridge, South of Venoco bridge, Main Channel, South of Phelps bridge, North of Phelps bridge, Phelps Road, and Devereux Creek. Data points are shown as open circles (unexcavated) and filled blue circles (excavated).

Site	Excavated (log + 1 transformed average abundance/L)	Unexcavated (log + 1 transformed average abundance/L)
North of Venoco bridge	0.0	0.1, 1.8
South of Venoco bridge	0.0	0.2, 1.2
Main Channel	0.4	0.0, 0.1, 1.1
South of Phelps bridge	2.3	0.8, 0.9, 1.3, 1.4, 1.7, 3.0, 4.3, 4.4, 5.3, 6.4
North of Phelps bridge	0.0	0.4, 1.1
Phelps Road	0.3	0.0, 0.1, 1.1
Devereux Creek	1.9, 2.0	0.0, 1.3, 2.3, 4.0

e. Interpretation of the figure

The abundance of copepods north and south of Venoco Bridge and at Phelps Road is higher than the abundance of ostracods, the abundance of each is similar in the Main Channel, south of Phelps Bridge, north of Phelps Bridge, and at Devereux creek. There does not seem to be a strong correlation between of the copepods and ostracods with the salinity.

2.

a.

- X axis: salinity (med_sal)
- Y-axis: log transformed average abundance/L (log_ave_lit)
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~taxon, scales = "free"), aq_ins_clean

b. Wrangle data

```
abundance_salinity <- aq_ins_clean |>
# log + 1 transform average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1)) |>
# convert taxon to title case
mutate(taxon = str_to_title(taxon)) |>
# order taxon levels by maximum abundance
mutate(taxon = fct_reorder(taxon, ave_lit, .fun = "max"))

# display 10 random rows
slice_sample(abundance_salinity, n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022-02-25_NEC	2022-02-25 00:00:00	NEC	Anne~	0	8.75	44.1	9.87
2	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	Clad~	264.	7.34	4.1	6.1
3	2022-08-12_NEC	2022-08-12 00:00:00	NEC	Dipt~	0	7.37	NA	NA
4	2022-08-10_NPB	2022-08-10 00:00:00	NPB	Dipt~	0	8.28	8.84	9.81
5	2022-02-25_NEC	2022-02-25 00:00:00	NEC	Cope~	52.3	8.75	44.1	9.87
6	2021-11-16_NPB	2021-11-16 00:00:00	NPB	Hemi~	0	7.3	2.38	12.1
7	2022-08-15_NMC	2022-08-15 00:00:00	NMC	Anne~	0	7.61	NA	NA

```

8 2023-01-28_NPB1 2023-01-28 00:00:00 NPB1 Dipt~ 0.533 7.66 NA 9.74
9 2023-05-16_NVBR 2023-05-16 00:00:00 NVBR Clad~ 0 9.29 NA 9.83
10 2021-11-16_NPB 2021-11-16 00:00:00 NPB Clad~ 0 7.3 2.38 12.1
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, log_ave_lit <dbl>

```

c.

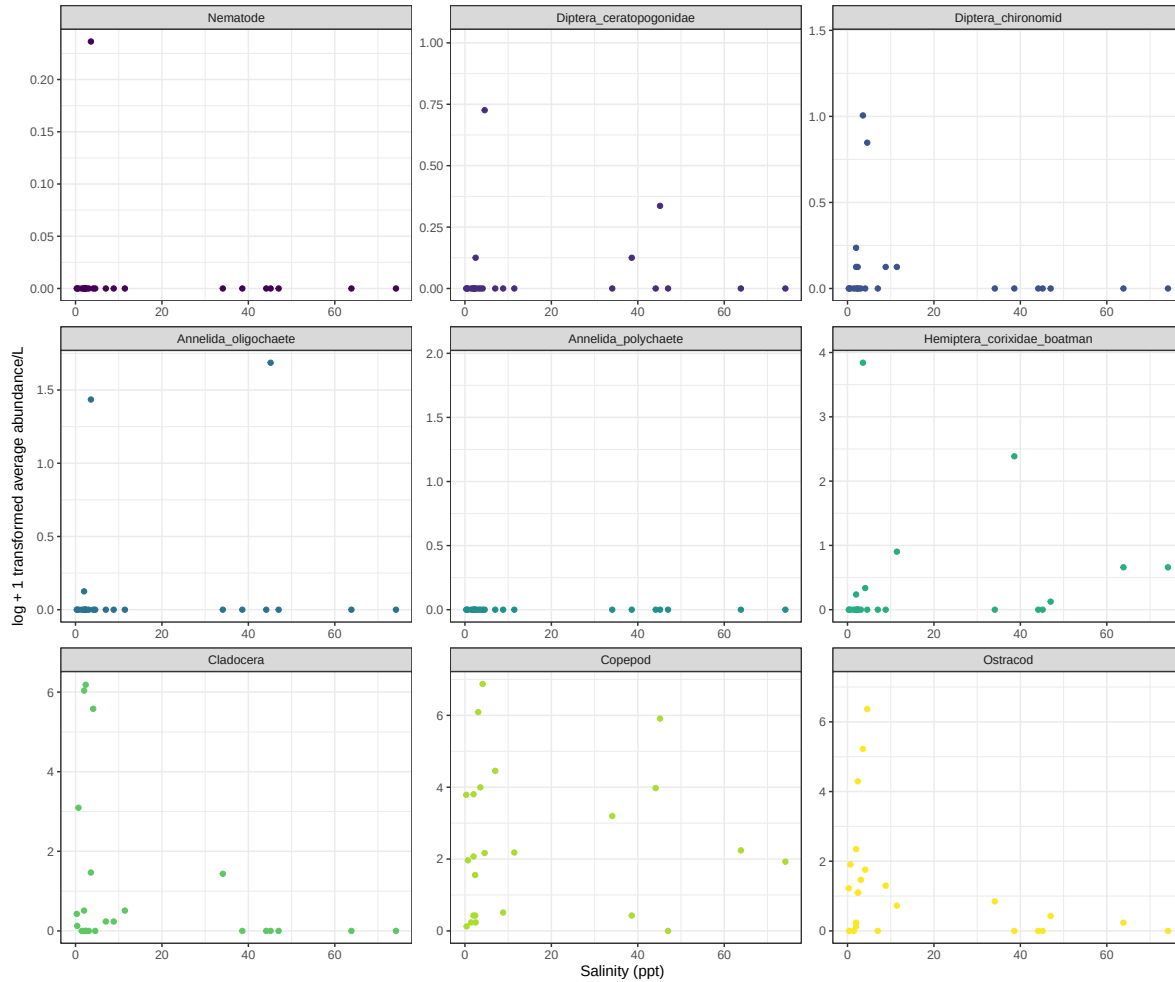
```

abundance_salinity_plot <- ggplot(data = abundance_salinity,
                                   mapping = aes(x = med_sal,
                                                  y = log_ave_lit,
                                                  color = taxon)) +

  # points for each observation
  geom_point() +
  # separate panel for each taxon with free scales
  facet_wrap(~taxon, scales = "free") +
  # readable axis labels
  labs(x = "Salinity (ppt)",
       y = "log + 1 transformed average abundance/L") +
  # different theme
  theme_bw() +
  # different colors
  scale_color_viridis_d() +
  # remove legend
  theme(legend.position = "none")

abundance_salinity_plot

```



d. Interpreting figure

Nematodes spike about a salinity of 5ppt but stay around 0 for every other salinity value; a similar pattern is seen in the annelida oligochaete where is spikes at arond 5ppt and 45ppt but is around 0 for every other value. Diptera ceratopogonidae has spikes around 5-10ppt and 30-50ppt. Cladocera is much more abundant at lower salinities ranging from 1-10ppt. The copepod graoh seems a bit more random/scattered but has higher abundance from 1-10ppt anf around 40ppt.